

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Adjustable Rail Motor Pulley & End Pulley Removal & Replacement	Effective Date	7 July 2017

This procedure is applicable to below models:

V810 STD	☐	S1	☐	S2
V810 XXL	☐	S1	☐	S2
V810 S2EX	☒			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	7 July 2017	First Release	Muhammed Shameem

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Adjustable Rail Motor Pulley & End Pulley Removal & Replacement	Effective Date	7 July 2017

Content

A. Adjustable Rail Motor Pulley Removal & Replacement.....	3
I. Adjustable Rail Motor Pulley Removal.....	3
II. Adjustable Rail Motor Pulley Replacement.....	6
B. Adjustable Rail End Pulley Removal & Replacement.....	13
I. Adjustable Rail End Pulley Removal.....	13
II. Adjustable Rail End Pulley Replacement.....	14
C. Adjustable Rail Belt Alignment.....	18

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Adjustable Rail Motor Pulley & End Pulley Removal & Replacement	Effective Date	7 July 2017

A. Adjustable Rail Motor Pulley Removal & Replacement

I. Adjustable Rail Motor Pulley Removal

1. Turn Off X-ray Source and open Front Access Door
2. Unplug Adjustable Rail Conveyor Motor Connector as shown in Figure 1

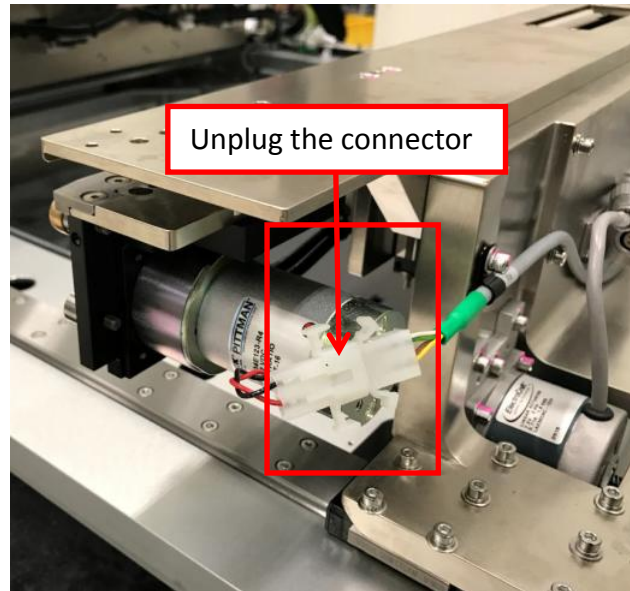


Figure 1 Adjustable Rail Conveyor Motor Connector

3. Loosen 1 x Screw at Tension Pulley Kit as shown in Figure 2.
Gently push Tension Pulley Kit towards left direction (Blue arrow in Figure 2)
Remove Conveyor Belt from Adjustable Rail

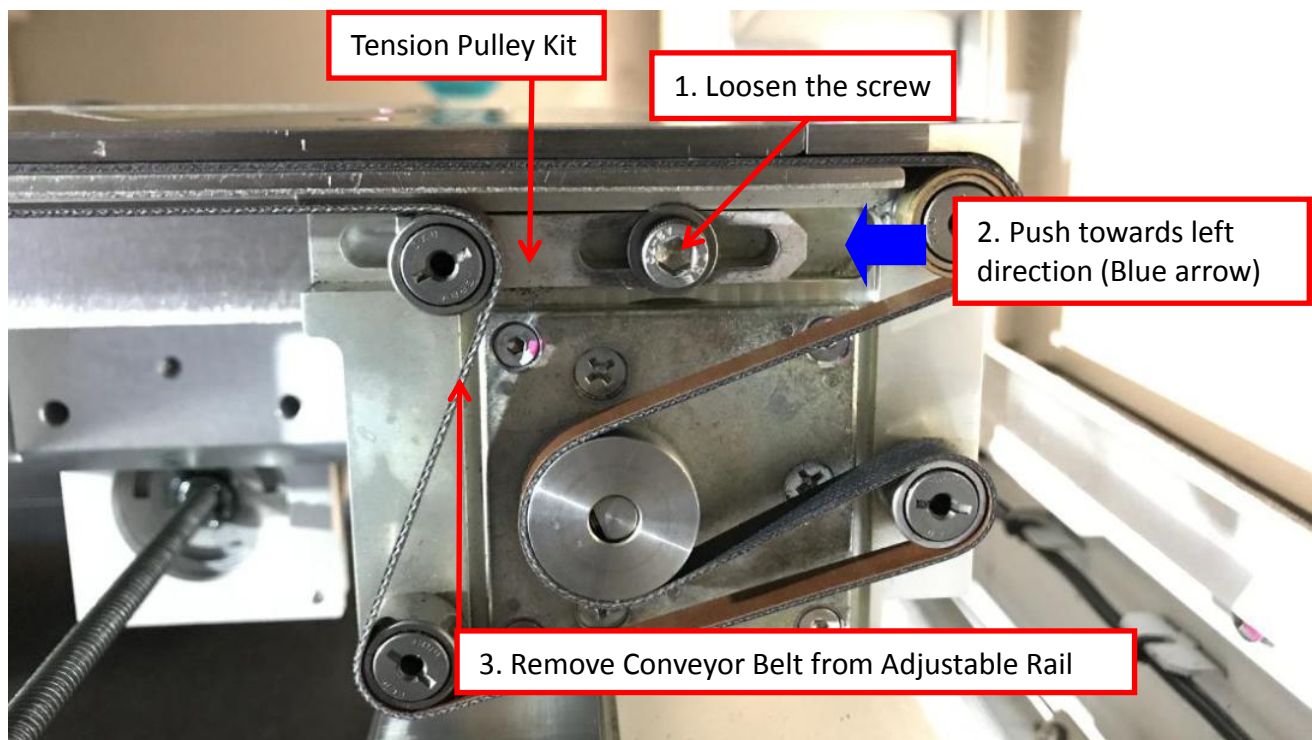


Figure 2 Adjustable Rail Conveyor Belt

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Adjustable Rail Motor Pulley & End Pulley Removal & Replacement	Effective Date	7 July 2017

4. Loosen 4 x Screws. Remove Adjustable Rail Motor Pulley as shown in Figure 3 & 4

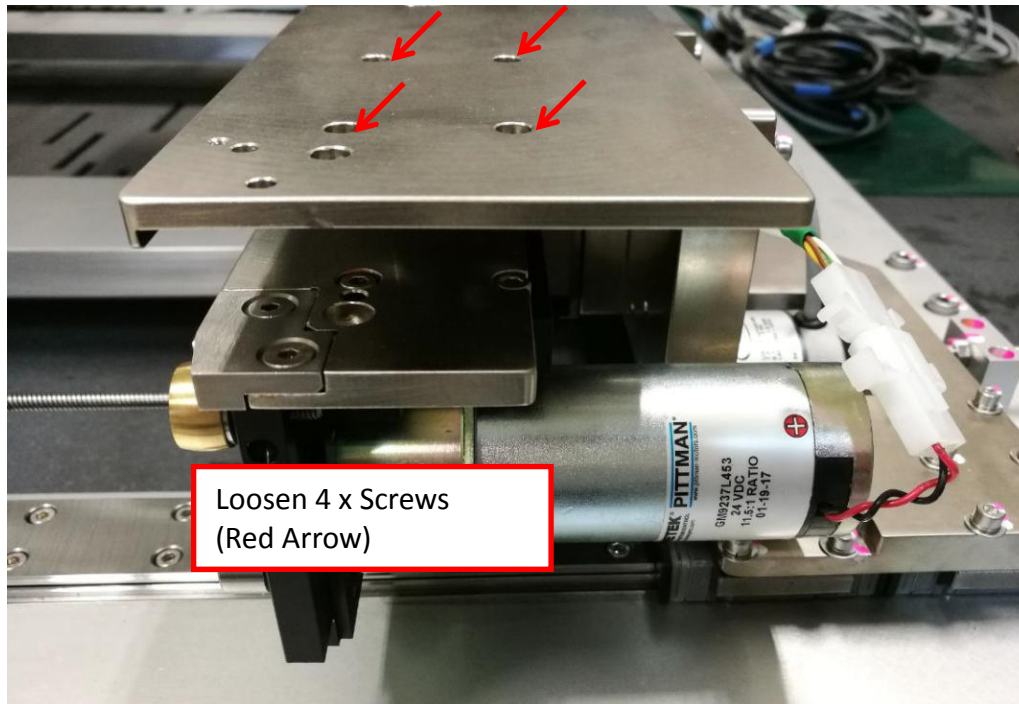


Figure 3 Adjustable Rail Motor Pulley

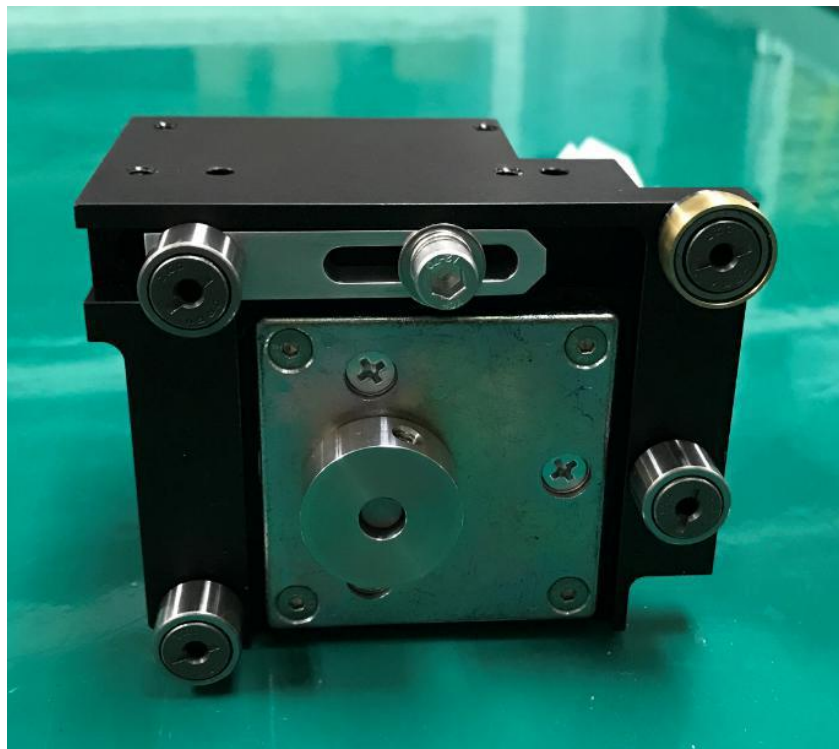


Figure 4 Adjustable Rail Motor Pulley

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Adjustable Rail Motor Pulley & End Pulley Removal & Replacement	Effective Date	7 July 2017

5. Loosen 1 x Screw from Tension Pulley Kit as Figure 5. Remove Tension Pulley Kit & Conveyor Motor as shown in Figure 5 to Figure 7

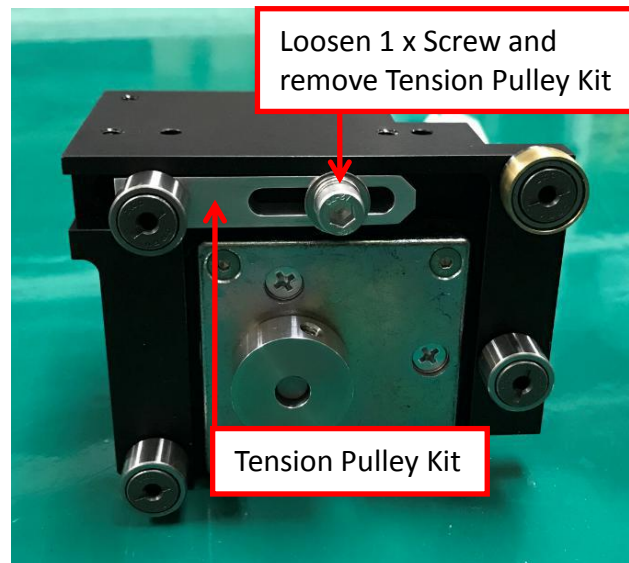


Figure 5 Tension Pulley Kit



Figure 6 Tension Pulley Kit

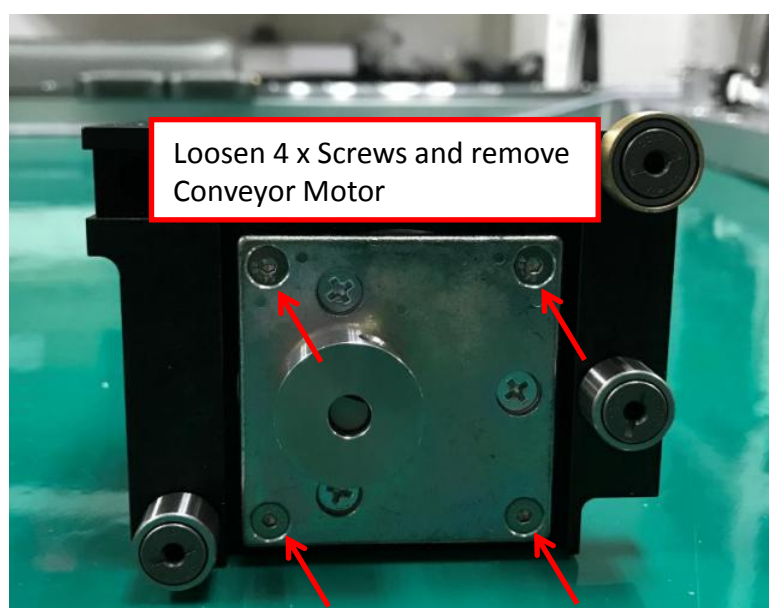


Figure 7 Conveyor Motor Removal

Remarks : Re-use item in Figure 5, 6 & 7

ViTrox	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Adjustable Rail Motor Pulley & End Pulley Removal & Replacement	Effective Date	7 July 2017

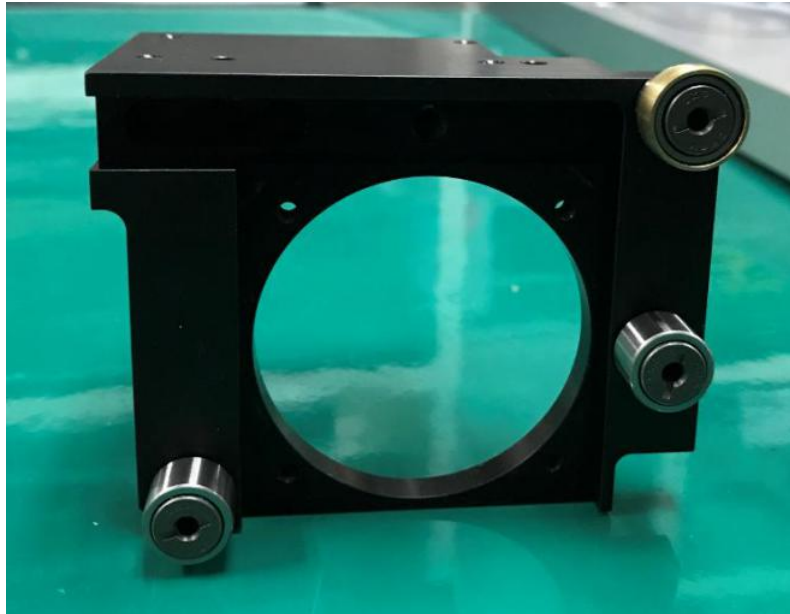


Figure 8 Adjustable Rail Motor Pulley (After Conveyor Motor Removal)

II. Adjustable Rail Motor Pulley Replacement

1. Prepare new Adjustable Rail Motor Pulley (Part ID: 4XASP-A141)
2. Install Conveyor Motor to “Adjustable Rail Motor Pulley” as shown in Figure 9 & 10

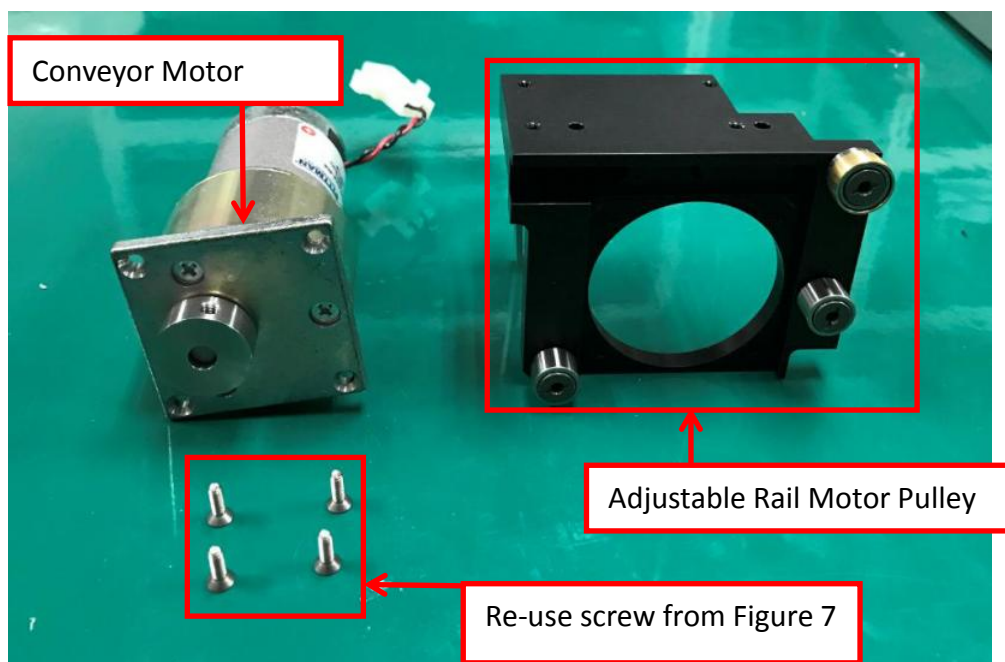


Figure 9 Conveyor Motor Assembly

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Adjustable Rail Motor Pulley & End Pulley Removal & Replacement	Effective Date	7 July 2017

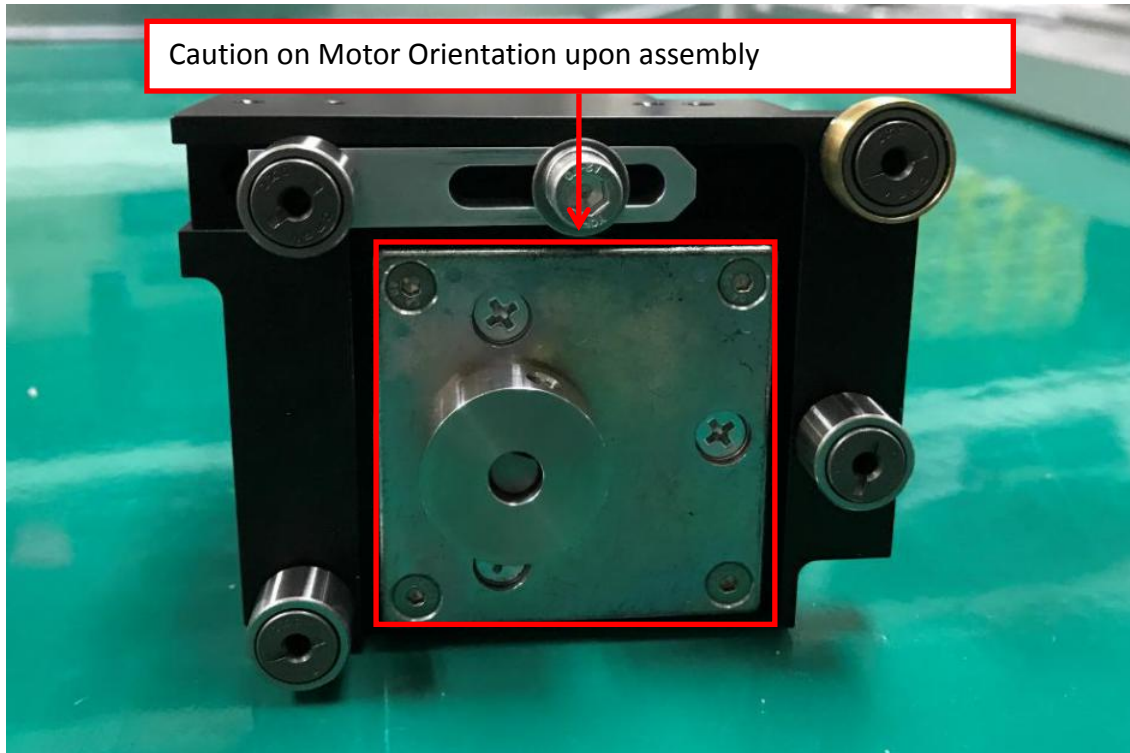


Figure 10 Conveyor Motor Assembly Orientation

3. Install Tension Pulley Kit as shown in Figure 11 & 12

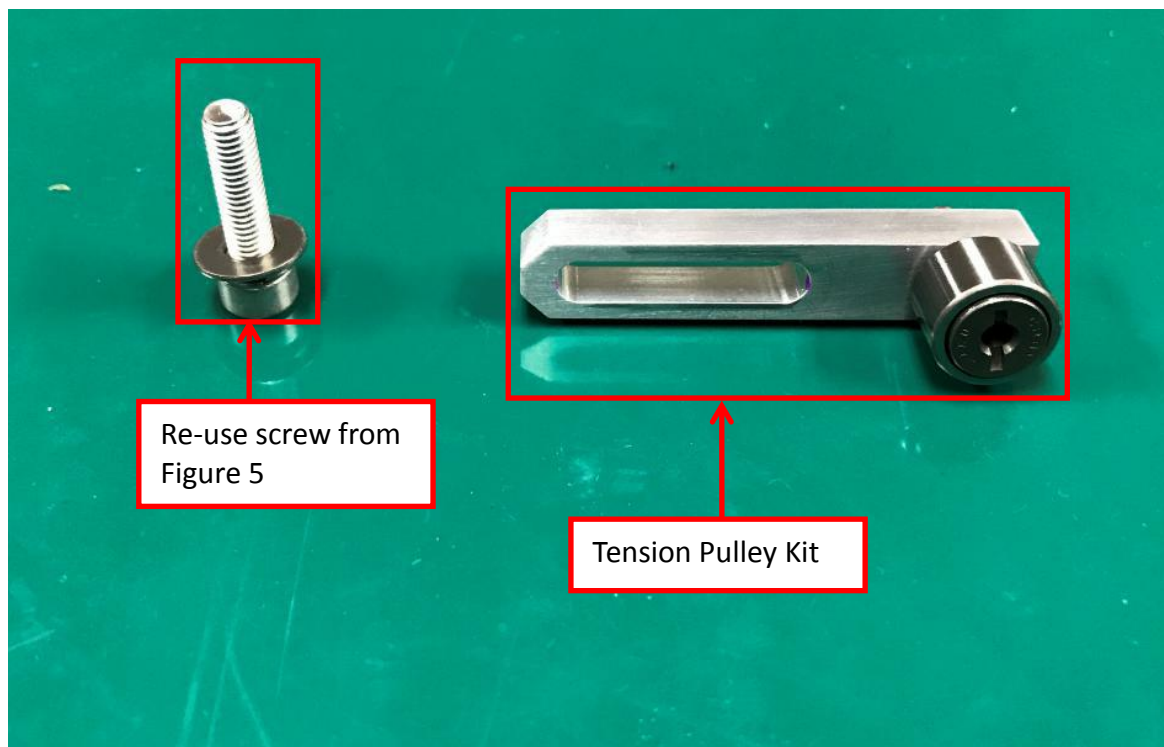


Figure 11 Tension Pulley Kit

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Adjustable Rail Motor Pulley & End Pulley Removal & Replacement	Effective Date	7 July 2017

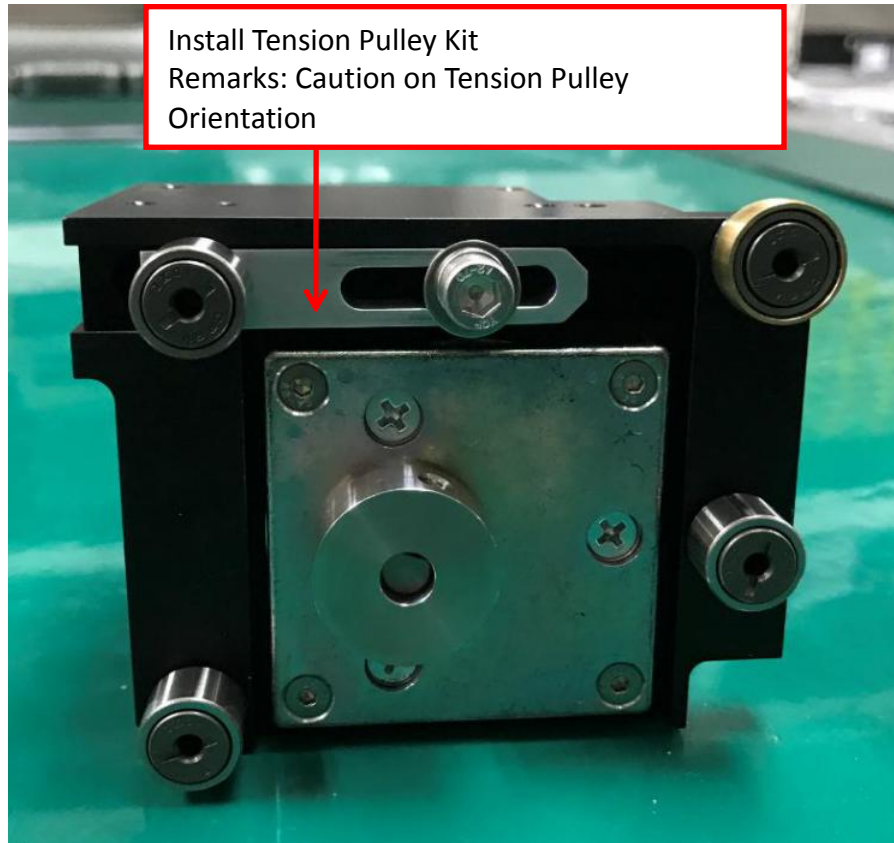


Figure 12 Tension Pulley Kit

4. Assemble Adjustable Rail Motor Pulley (Figure 13) into Adjustable Rail as shown from Figure 13 to 15

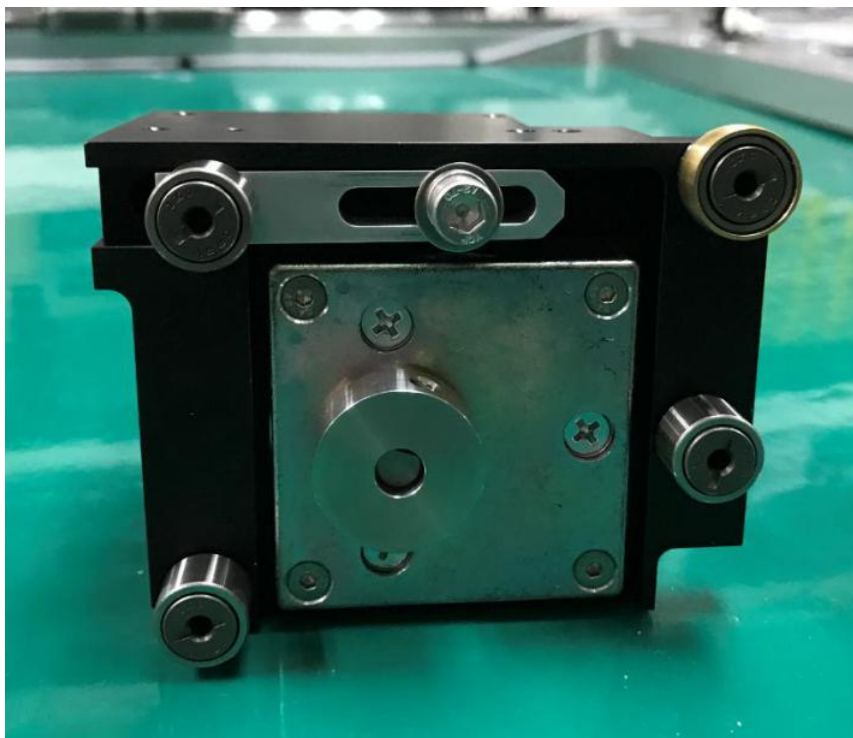


Figure 13 Adjustable Rail Motor Pulley (Full Assembly)

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Adjustable Rail Motor Pulley & End Pulley Removal & Replacement	Effective Date	7 July 2017

Install Adjustable Rail Motor Pulley

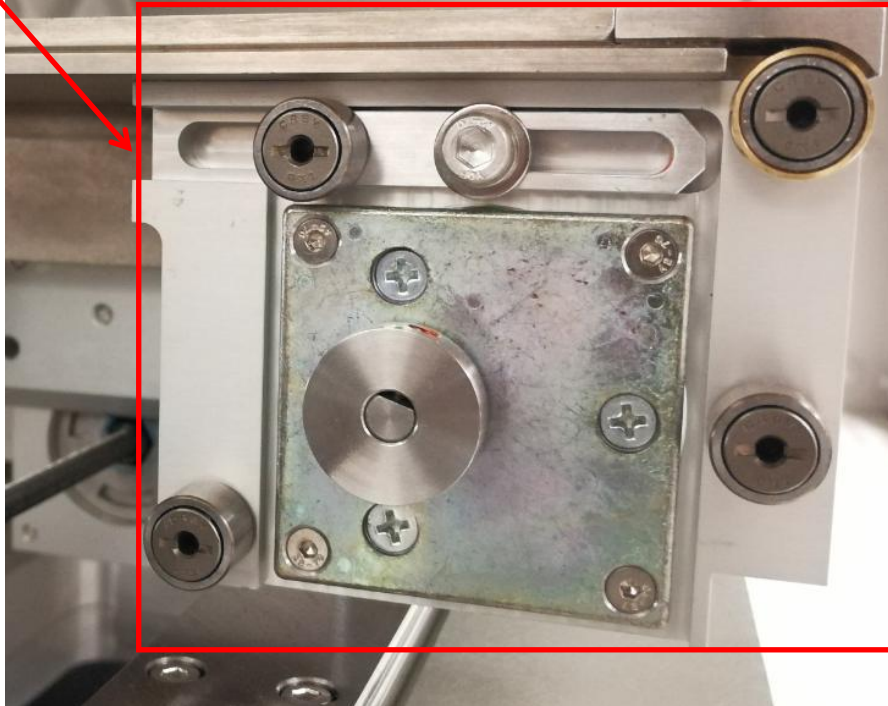


Figure 14 Adjustable Rail Motor Pulley

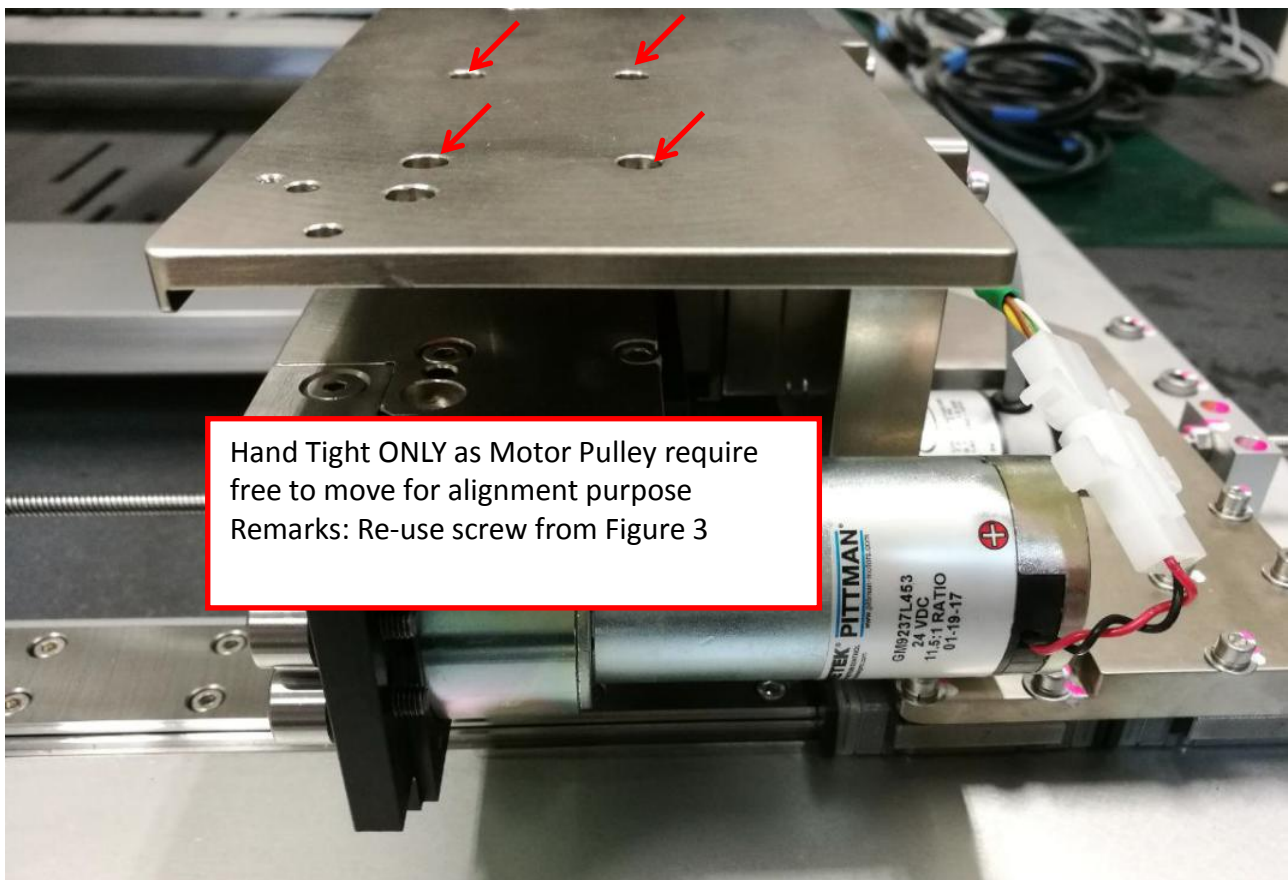


Figure 15 Adjustable Rail Motor Pulley Assembly

ViTrox	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Adjustable Rail Motor Pulley & End Pulley Removal & Replacement	Effective Date	7 July 2017

5. Install Conveyor Belt to the rail by refer to Figure 16. Hand tight Tension Pulley Kit

Figure 16 indicate Belt Routing Diagram

Figure 17 and 18 indicate Physical Routing at Motor Pulley & End Pulley



Figure 16 Routing Diagram

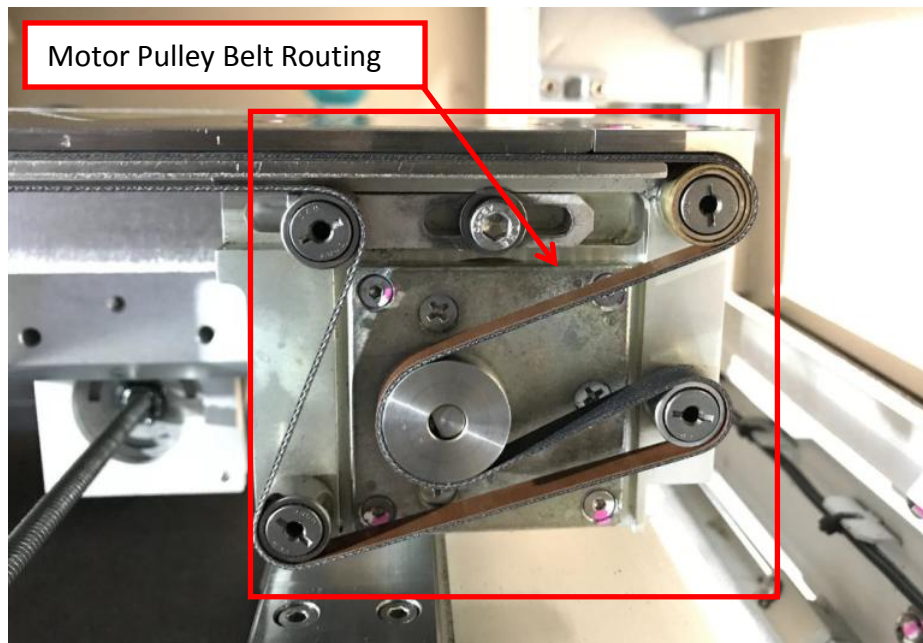


Figure 17 Motor Pulley Belt Routing

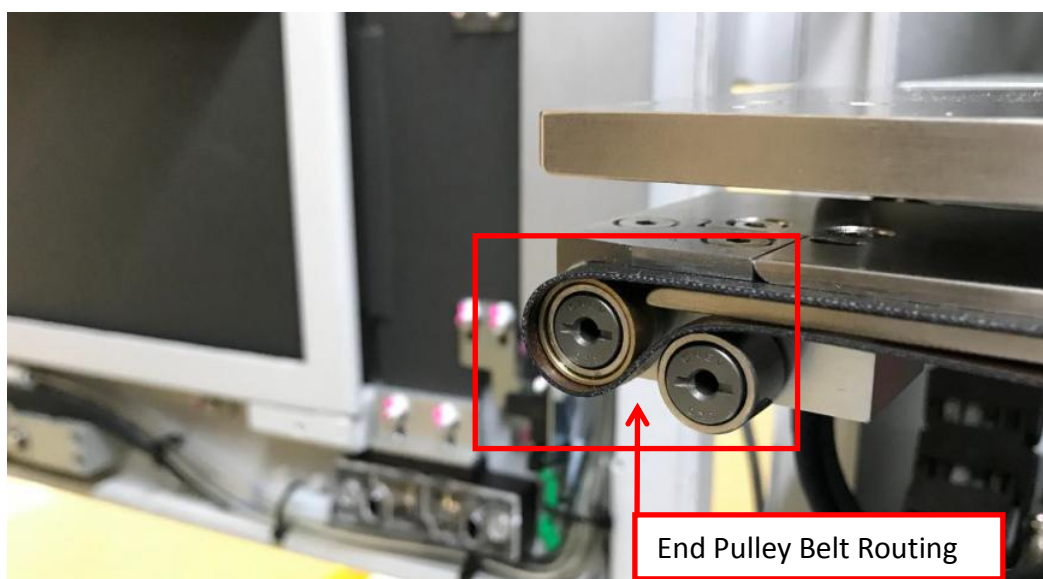


Figure 18 End Pulley Belt Routing

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Adjustable Rail Motor Pulley & End Pulley Removal & Replacement	Effective Date	7 July 2017

6. Prepare Belt Tension Kit (Part ID: 4XASP-A114) as shown in Figure 19

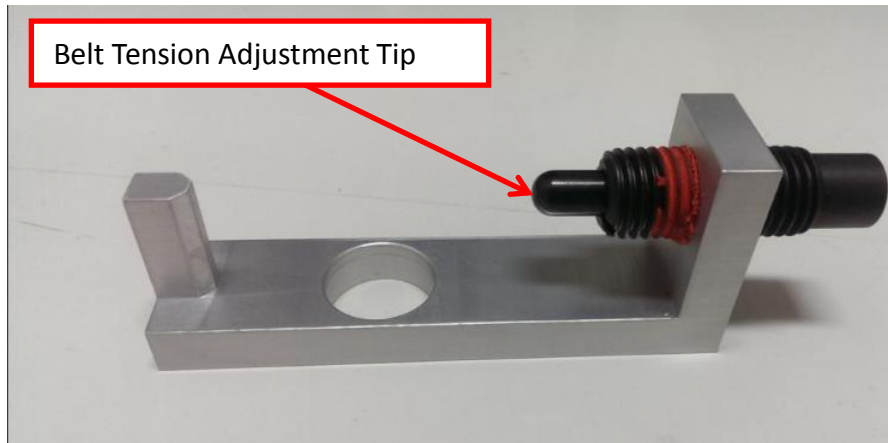


Figure 19 Belt Tension Kit

7. Gently rest Belt Tension Adjustment Tip on Tiny Opening as below Figure 20 ,while other end will require to hook with the Tension Pulley as Figure 21

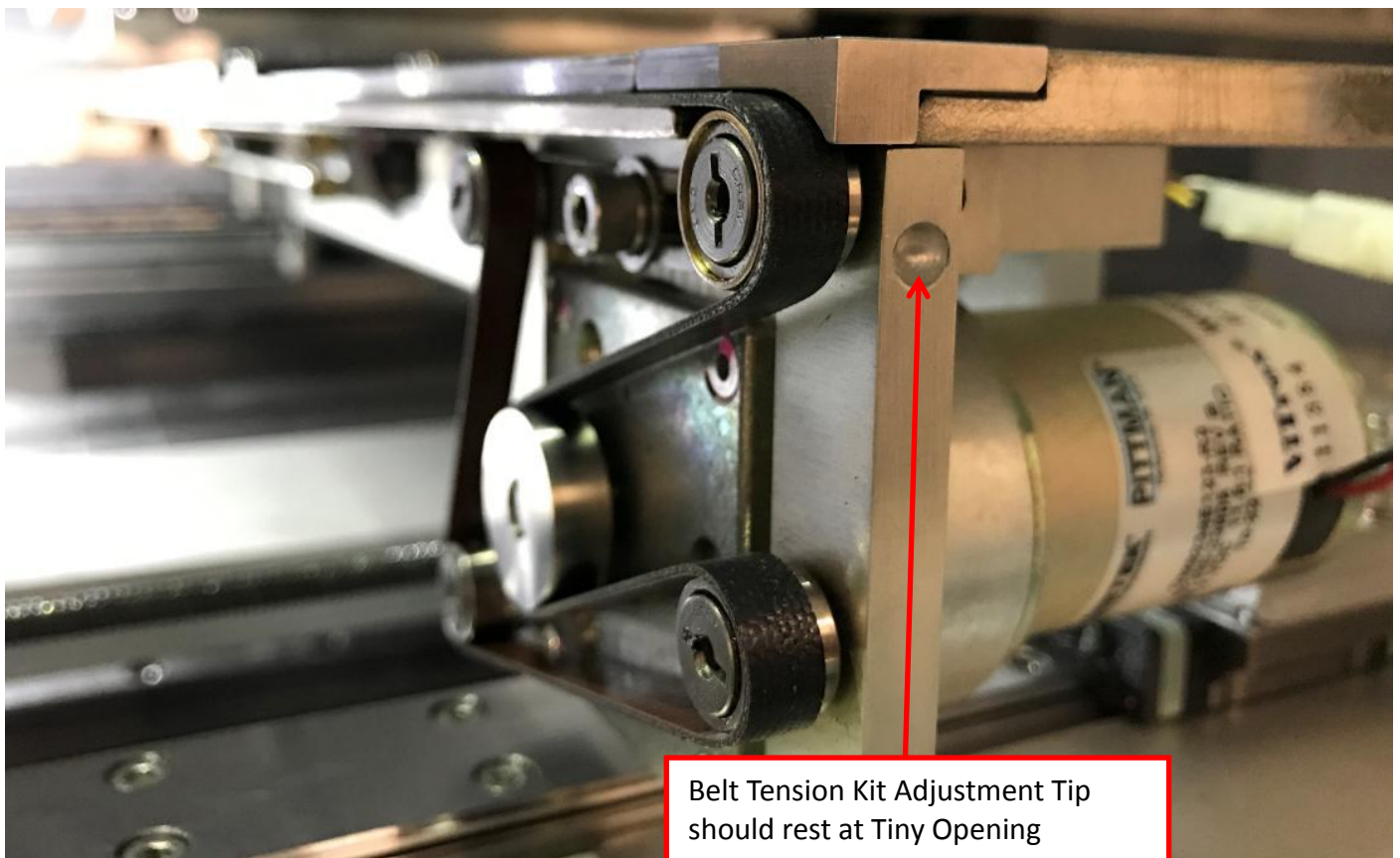


Figure 20 Belt Tension Kit Placement

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Adjustable Rail Motor Pulley & End Pulley Removal & Replacement	Effective Date	7 July 2017

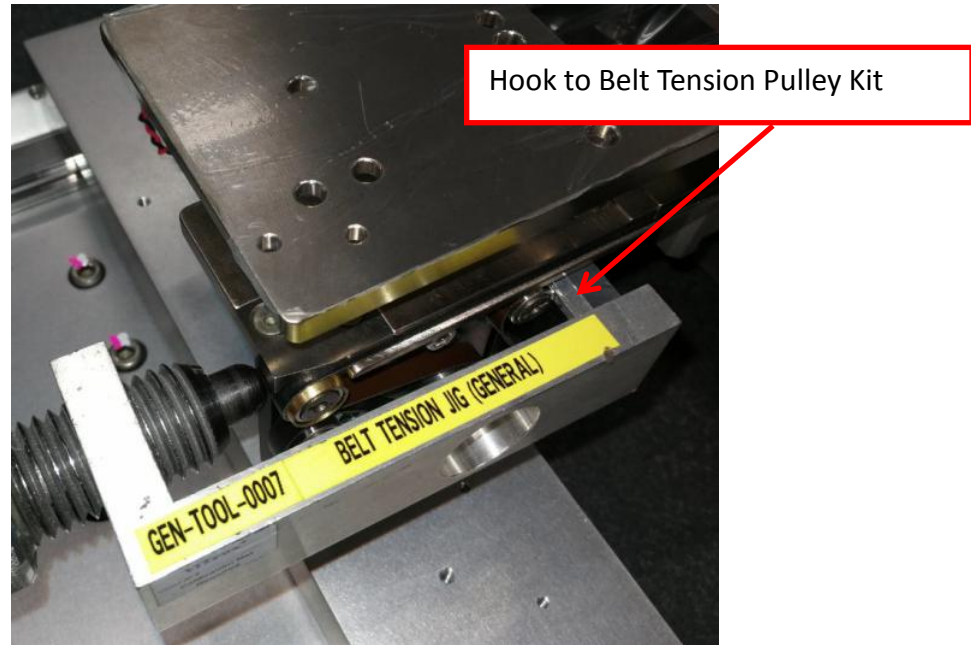


Figure 21 Belt Tension Kit

8. Loosen the Belt Tension Pulley Block Screw. Gently adjust Adjustment Knob until Tension Block is within 3/4 adjustment scale. Then, tighten the Belt Tension Pulley Block Screw

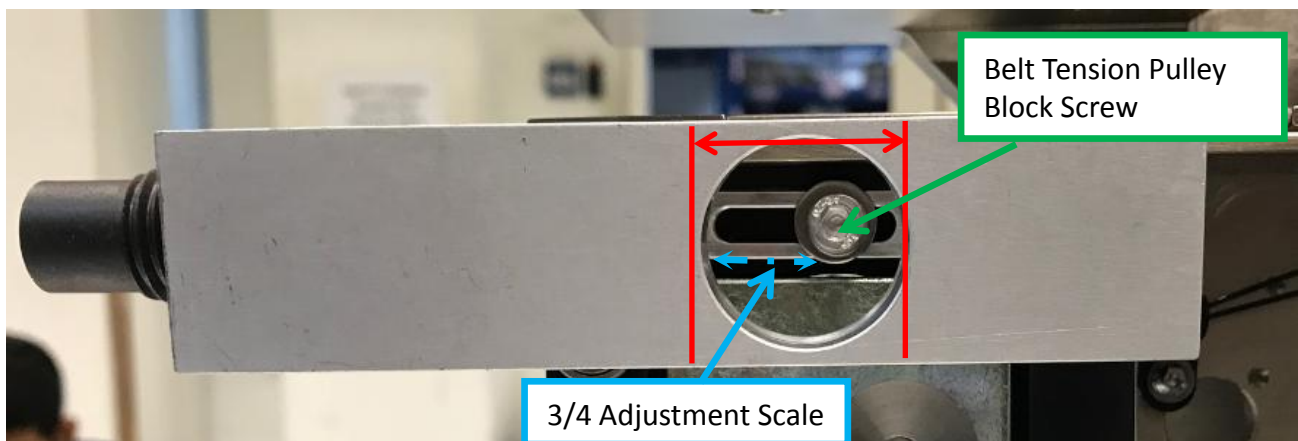


Figure 22 Belt Pulley Kit Tensioning

9. Remove 4XASP-A114 from machine
10. Connect the Conveyor Motor Connector (Refer to Figure 1)
11. May refer to Section C for Belt Alignment Procedure

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Adjustable Rail Motor Pulley & End Pulley Removal & Replacement	Effective Date	7 July 2017

B. Adjustable Rail End Pulley Removal & Replacement

I. Adjustable Rail End Pulley Removal

1. Turn off X-ray Source and open Front Access Door
2. Loosen 1 x Screw at Tension Pulley Kit as shown in Figure 23.

Push the Tension Pulley Kit towards left direction (Blue arrow in Figure 23)

Remove Conveyor belt from Adjustable Rail

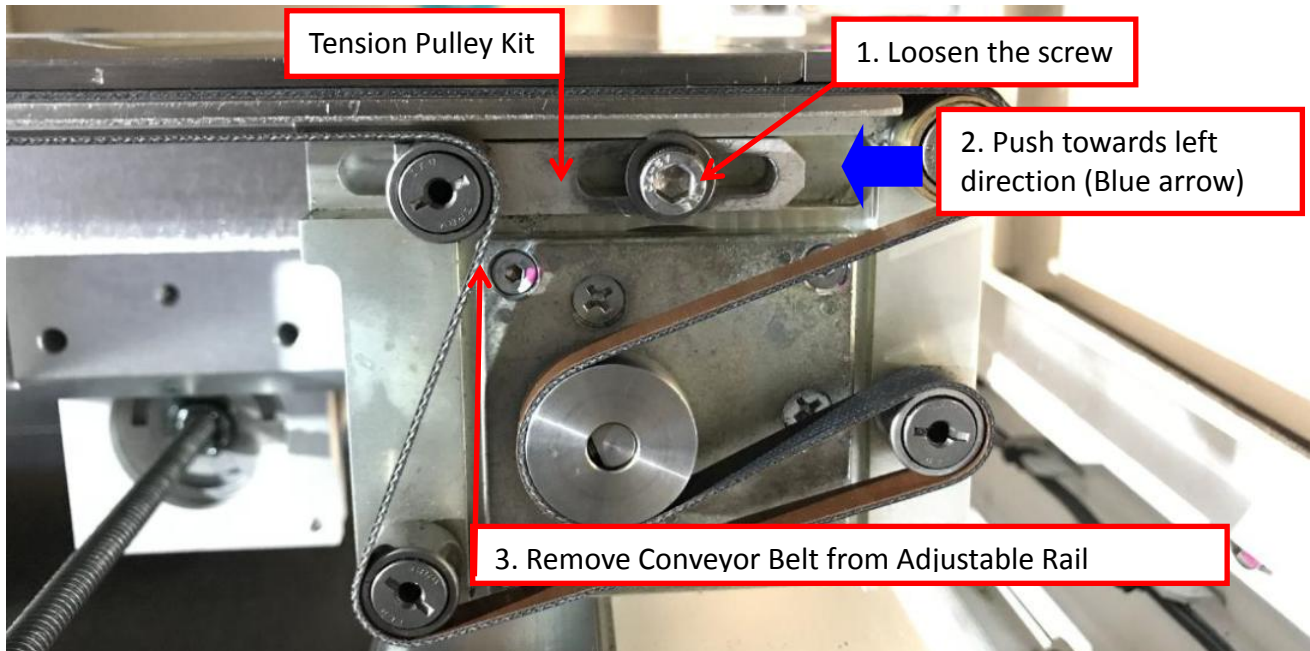


Figure 23 Adjustable Rail Conveyor Belt Removal

3. Loosen 2 x Screws. Remove End Pulley Kit from Adjustable Rail

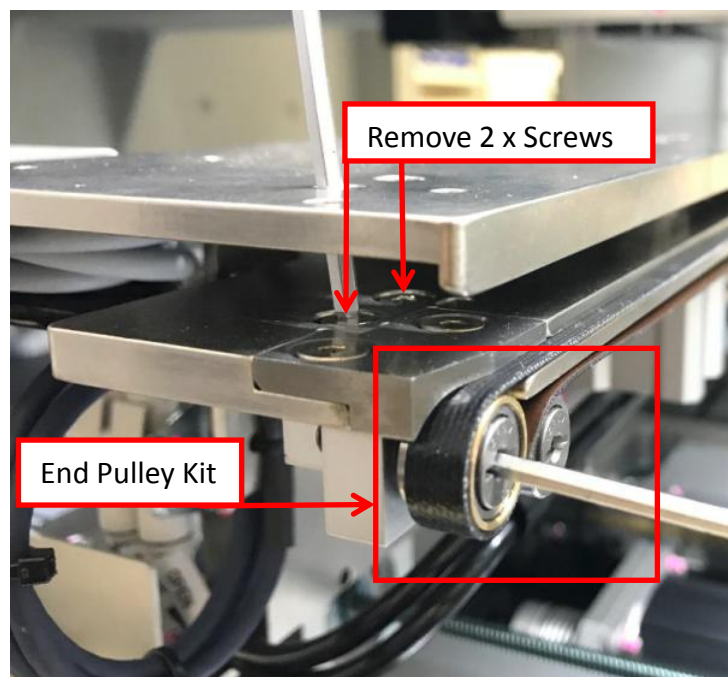


Figure 24 Adjustable Rail End Pulley Kit

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Adjustable Rail Motor Pulley & End Pulley Removal & Replacement	Effective Date	7 July 2017

II. Adjustable Rail End Pulley Replacement

1. Prepare 1 unit of 4XASP-A140 as shown in Figure 25

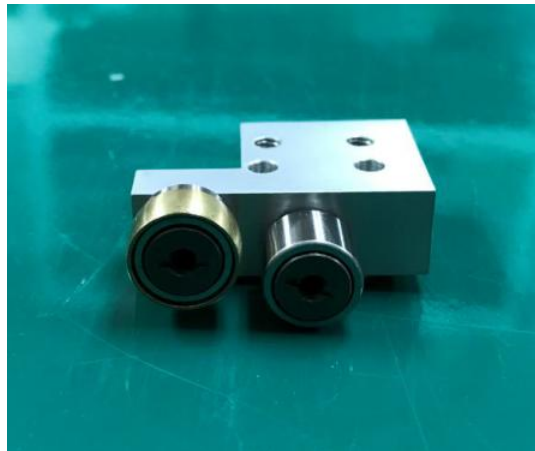


Figure 25 Adjustable Rail End Pulley Kit

2. Install Adjustable Rail End Pulley Kit into Adjustable Rail as shown in Figure 26. Tighten 2 x Screws at Adjustable Rail End Conveyor Pulley as Figure 27

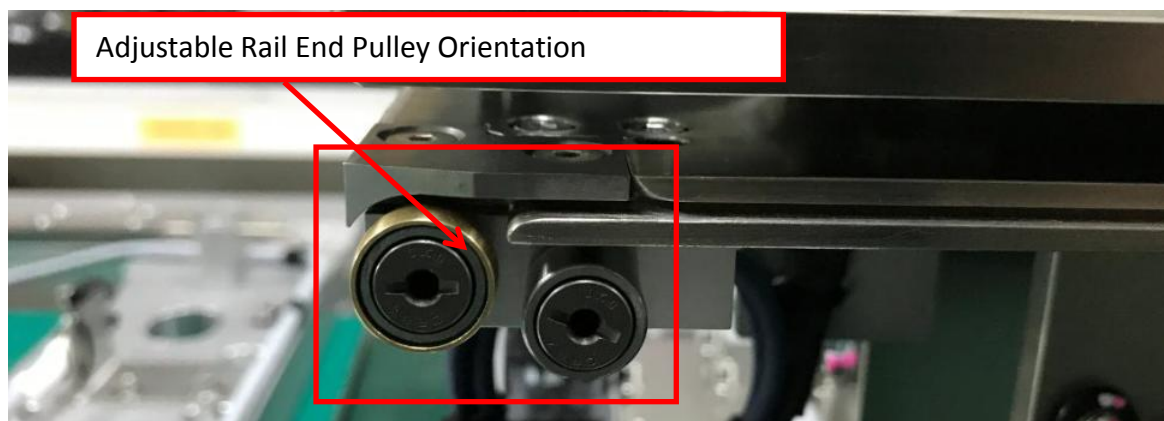


Figure 26 Adjustable Rail End Pulley Orientation

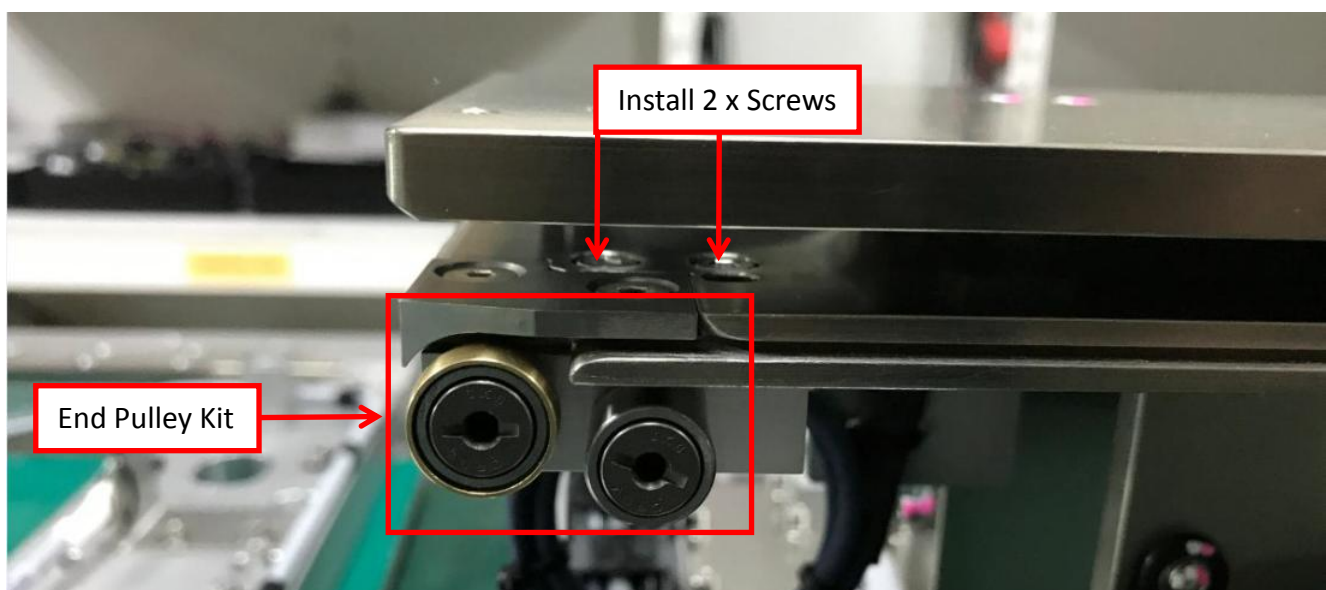


Figure 27 Adjustable Rail End Pulley Assembly

ViTrox	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Adjustable Rail Motor Pulley & End Pulley Removal & Replacement	Effective Date	7 July 2017

3. Install Conveyor Belt to the rail by refer to Figure 28. Hand tight Tension Pulley Kit

Figure 28 indicate Belt Routing Diagram

Figure 29 and 30 indicate Physical Routing at Motor Pulley & End Pulley



Figure 28 Routing Diagram

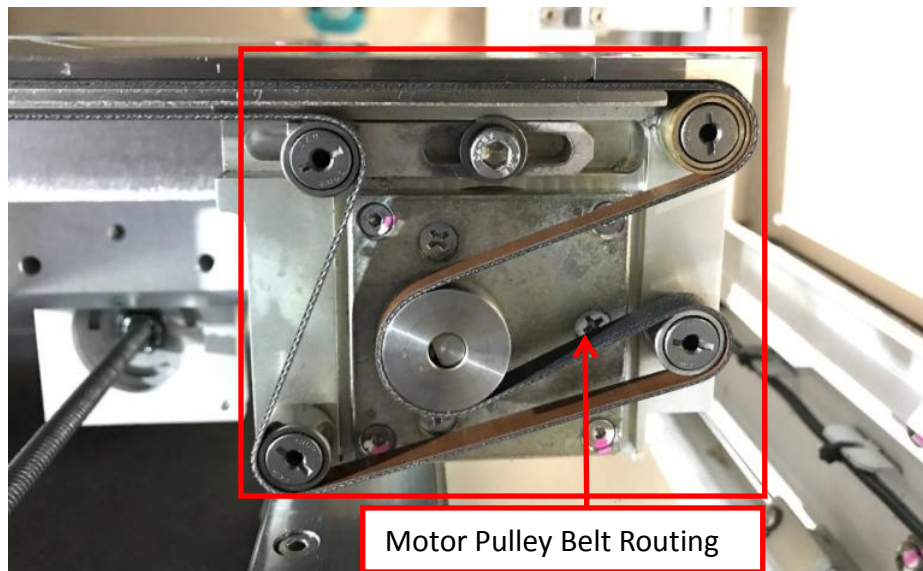


Figure 29 Motor Pulley Belt Routing

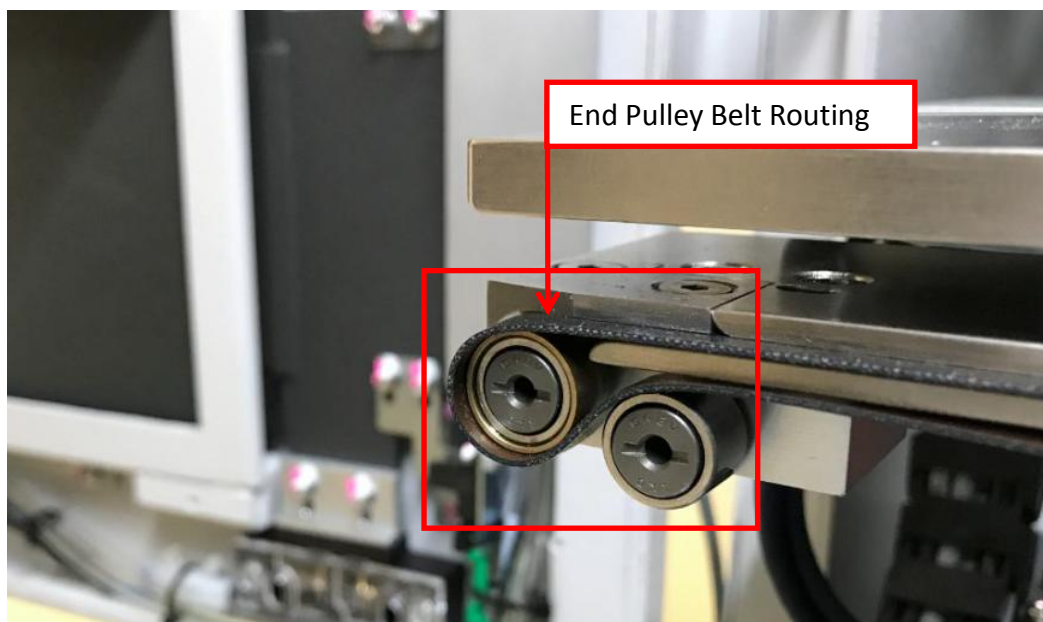


Figure 30 End Pulley Belt Routing

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Adjustable Rail Motor Pulley & End Pulley Removal & Replacement	Effective Date	7 July 2017

4. Prepare Belt Tension Kit (Part ID: 4XASP-A114) as shown in Figure 31

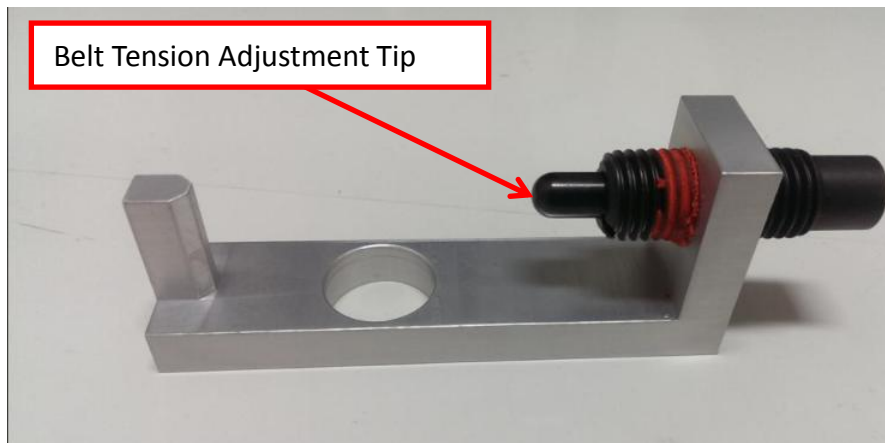


Figure 31 Belt Tension Kit

5. Gently rest Belt Tension Adjustment Tip on Tiny Opening as below Figure 32 ,while other end will require to hook with the Tension Pulley as Figure 33

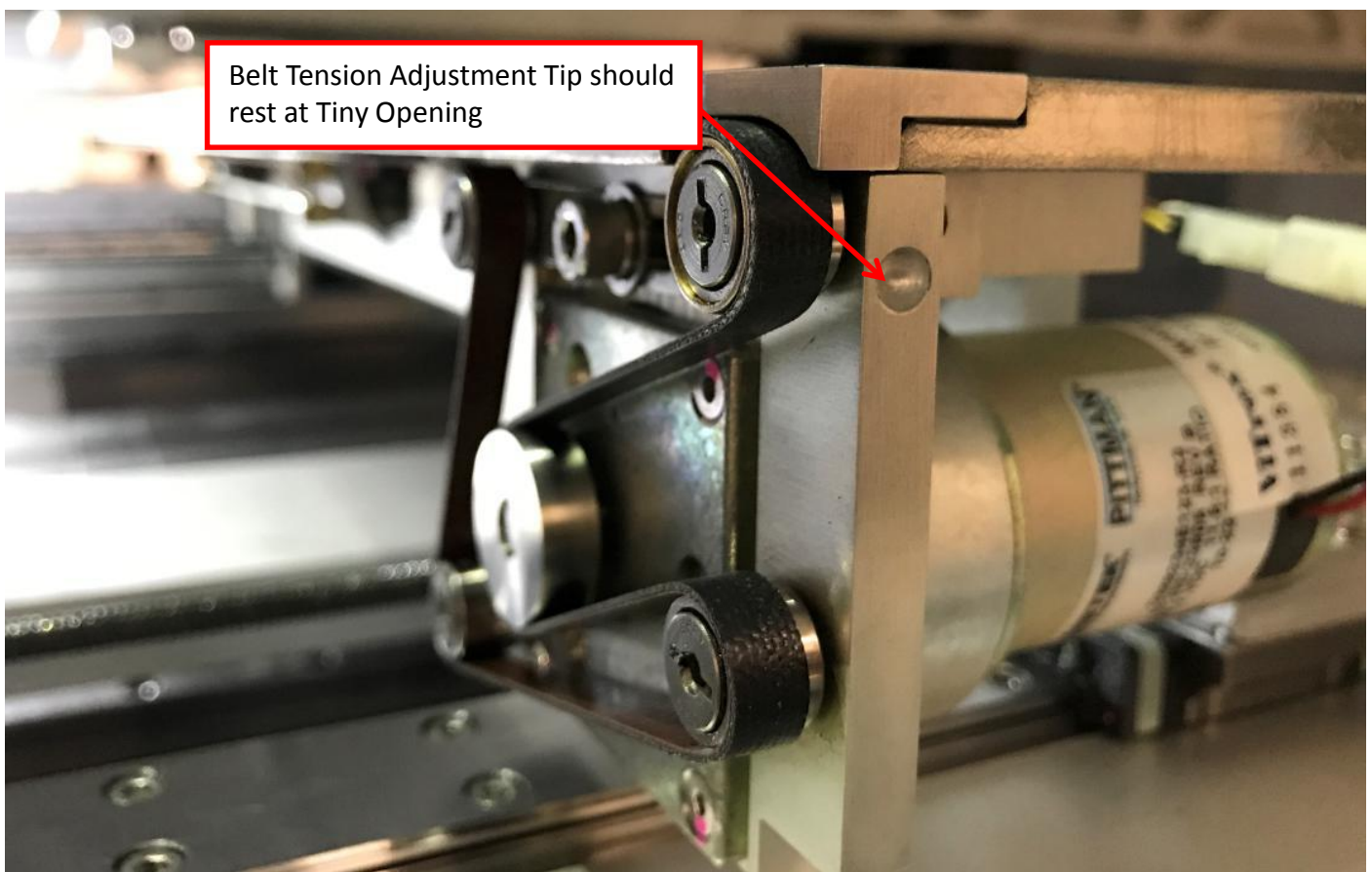


Figure 32 Belt Tension Kit Placement

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Adjustable Rail Motor Pulley & End Pulley Removal & Replacement	Effective Date	7 July 2017

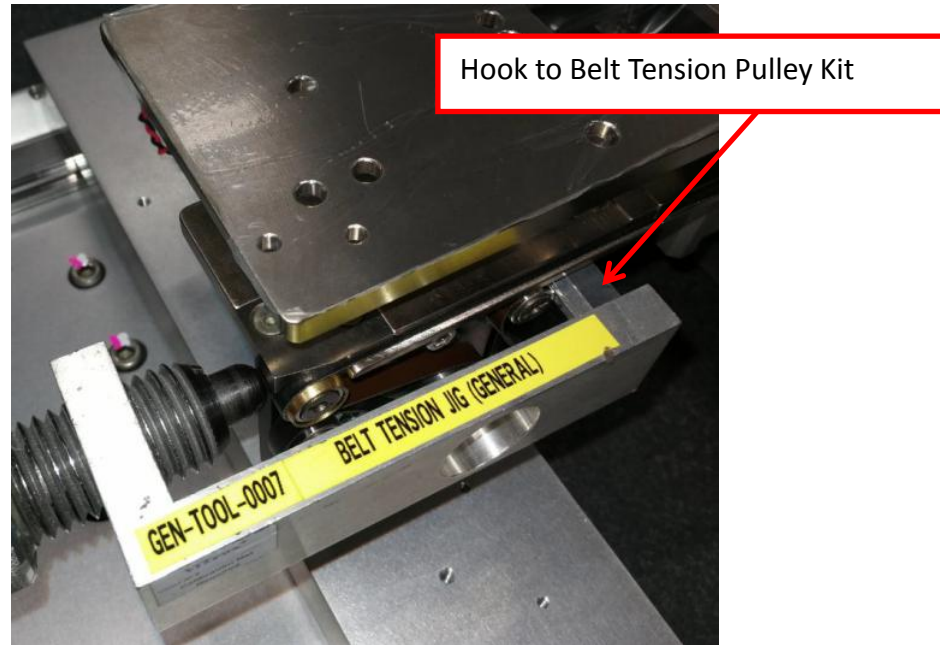


Figure 33 Belt Tension Kit

6. Loosen the Belt Tension Pulley Block Screw. Gently adjust Adjustment Knob until Tension Block is within 3/4 adjustment scale. Then, tighten the Belt Tension Pulley Block Screw

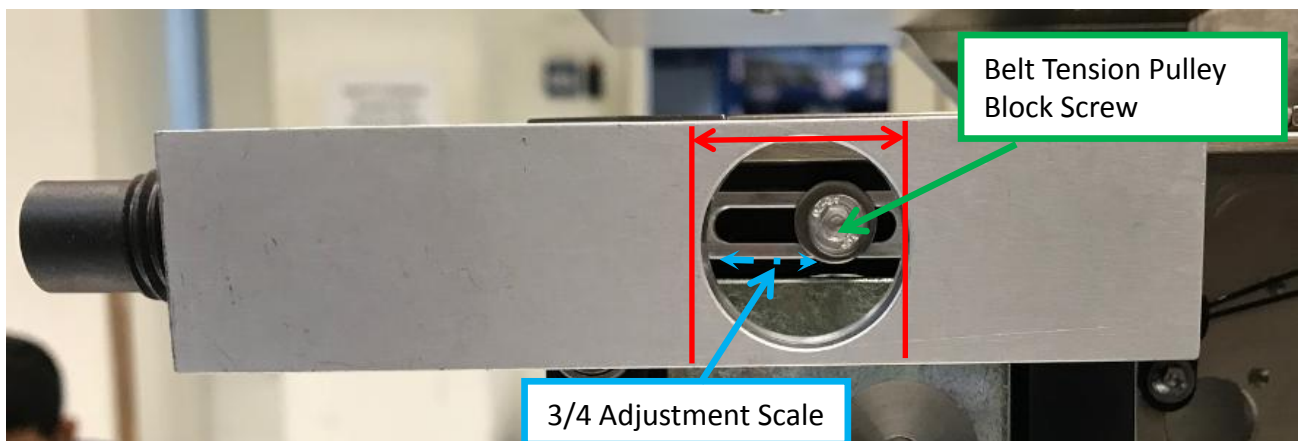


Figure 34 Belt Pulley Kit Tensioning

7. Remove 4XASP-A114 from machine
8. May refer to Section C for Belt Alignment Procedure

C. Adjustable Rail Belt Alignment

1. Turn Off X-Ray.
2. At V810 GUI, go to Services->Subsystem Status & Control->Digital IO and click “Initialize” as shown in Figure 35

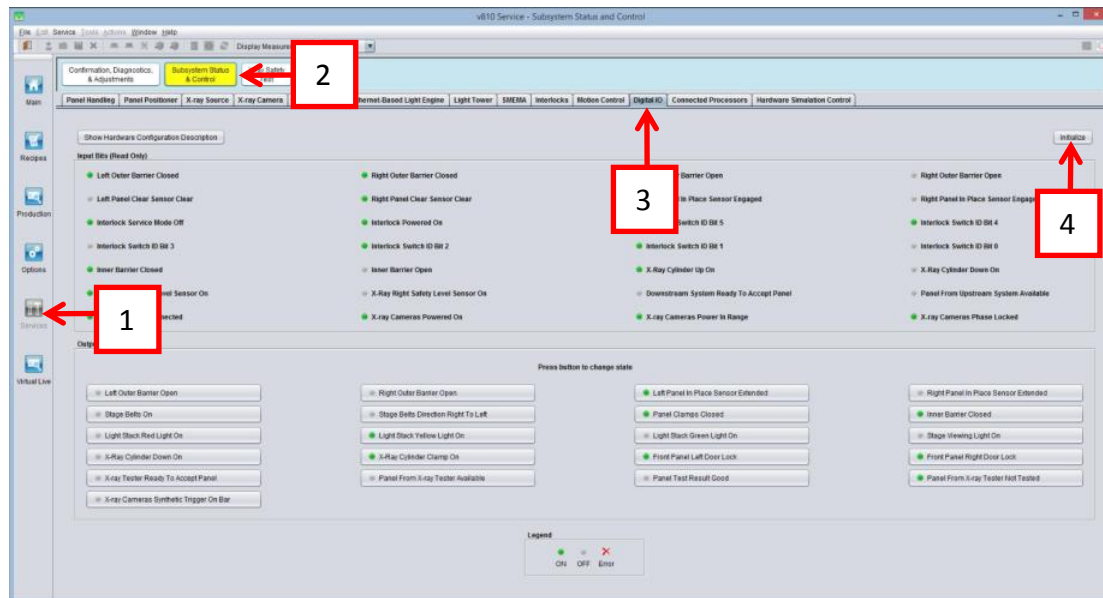


Figure 35 Digital IO

3. Toggle “Stage Belts On” as shown in Figure 36

Remarks : “Stage Belts Direction Right To Left” MUST Turn Off as shown in Figure 36.

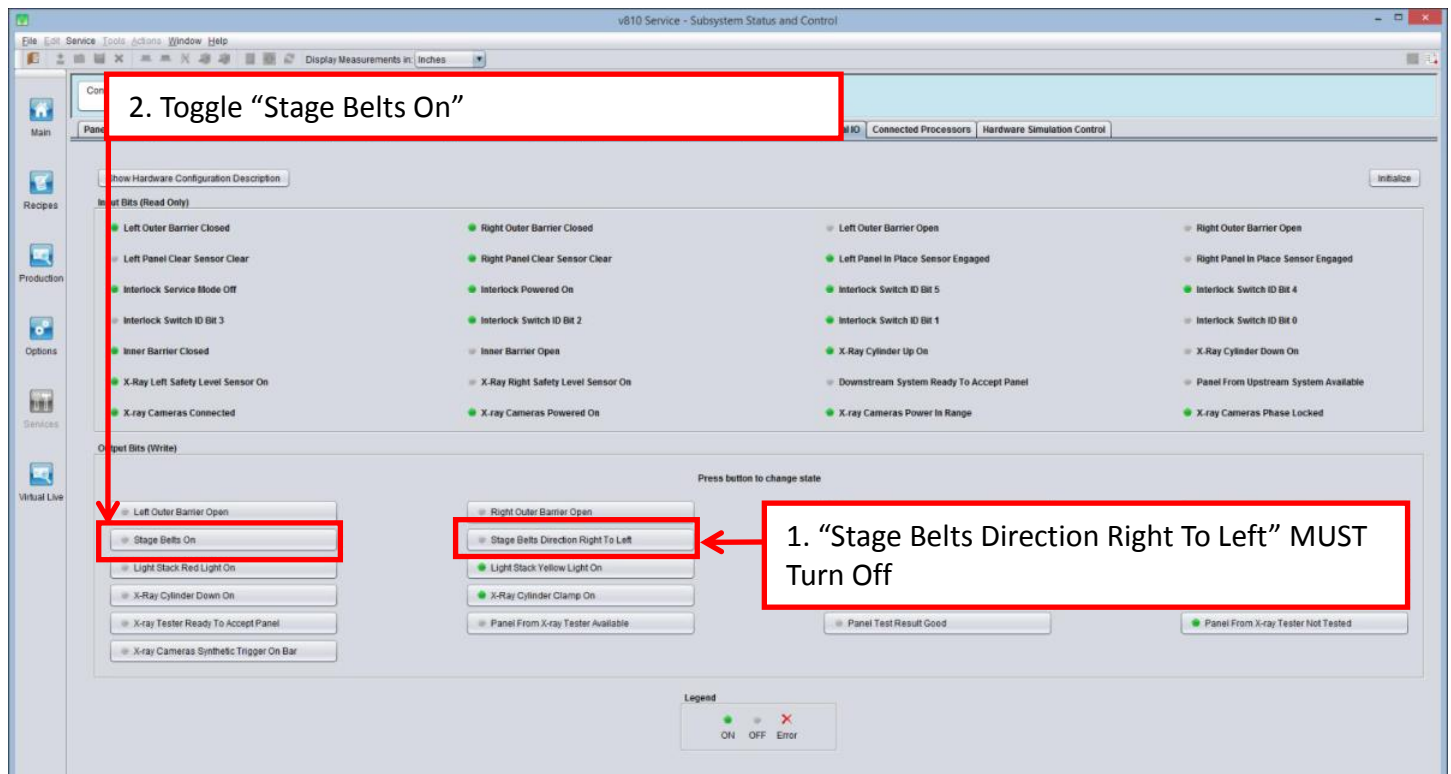


Figure 36 Digital IO

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Adjustable Rail Motor Pulley & End Pulley Removal & Replacement	Effective Date	7 July 2017

4. Open Front Access Door.
5. Visual check Conveyor Belt Running Condition by refer to Figure 37~42
 If Conveyor Belt Running Condition is Fail, may refer to Step 6 ~12,
 If Conveyor Belt Running Condition is Good, may refer to Step 7 ~12

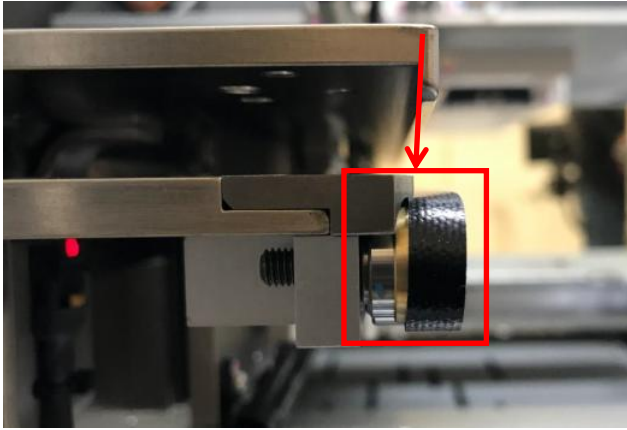


Figure 37 Belt Routing (NG)

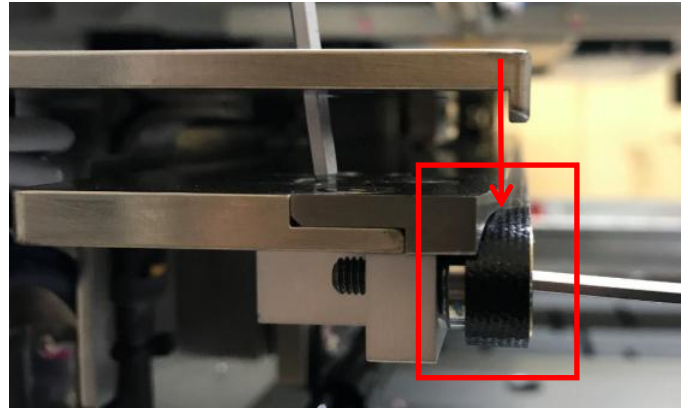


Figure 38 Belt Routing (GOOD)

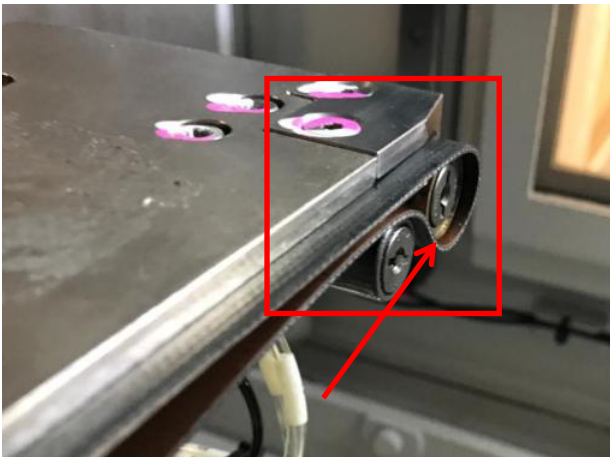


Figure 39 Belt Routing (NG)

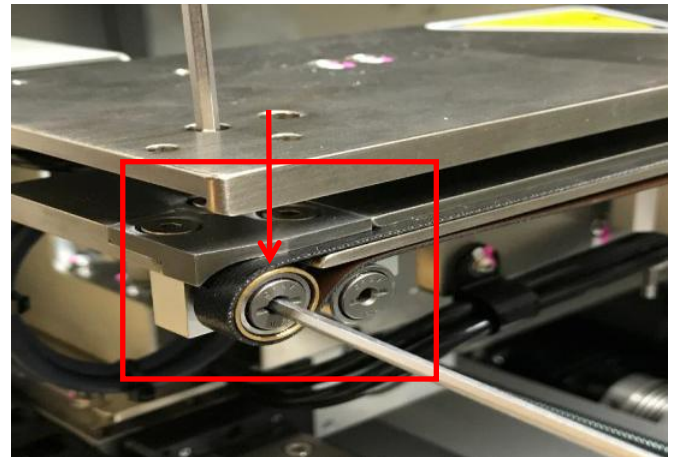


Figure 40 Belt Routing (GOOD)

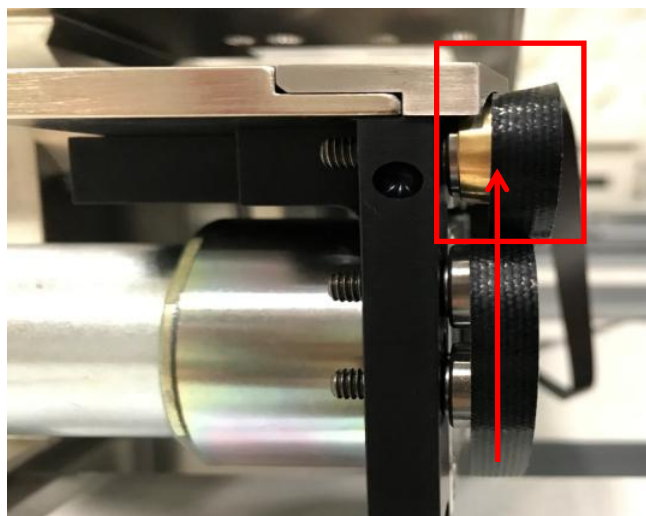


Figure 41 Belt Routing (NG)

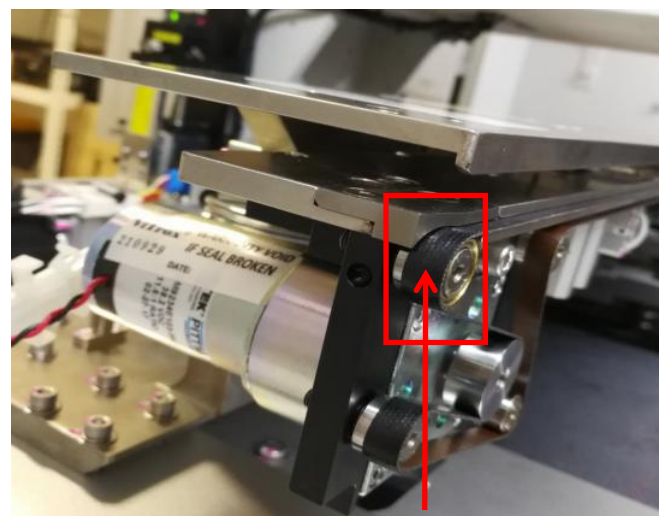


Figure 42 Belt Routing (GOOD)

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Adjustable Rail Motor Pulley & End Pulley Removal & Replacement	Effective Date	7 July 2017

6. Perform pulley alignment as below steps
 - a) Turn off the Belt Motor Motion at Digital IO Tab
 - b) Loosen 4 x Screws at Motor Pulley Assembly as below Figure 43

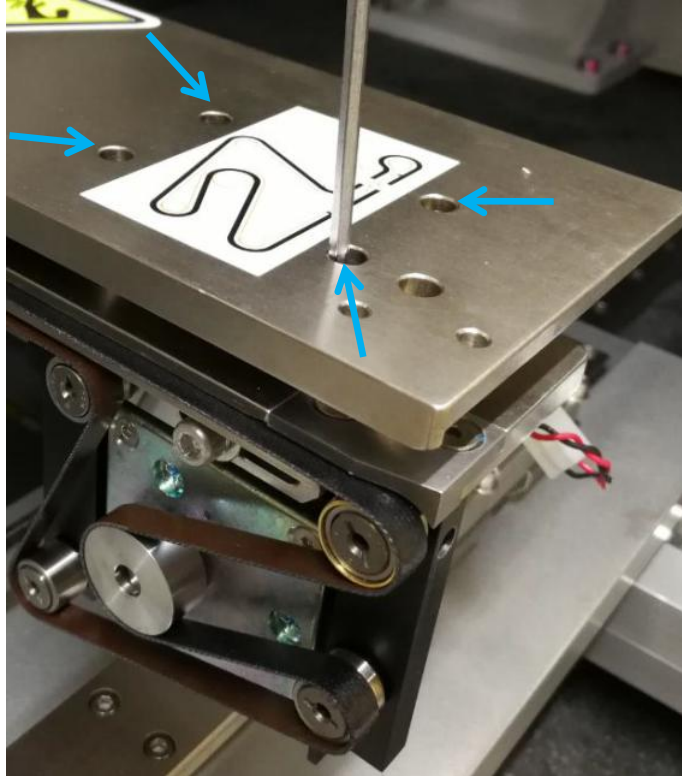


Figure 43 Adj Rail Mount Pulley Assembly

- c) Place L-ruler & Align Mount Pulley assembly as below Figure 44. Align Mount Pulley perpendicular with Adj Rail. Next, Tighten 4 x Screws

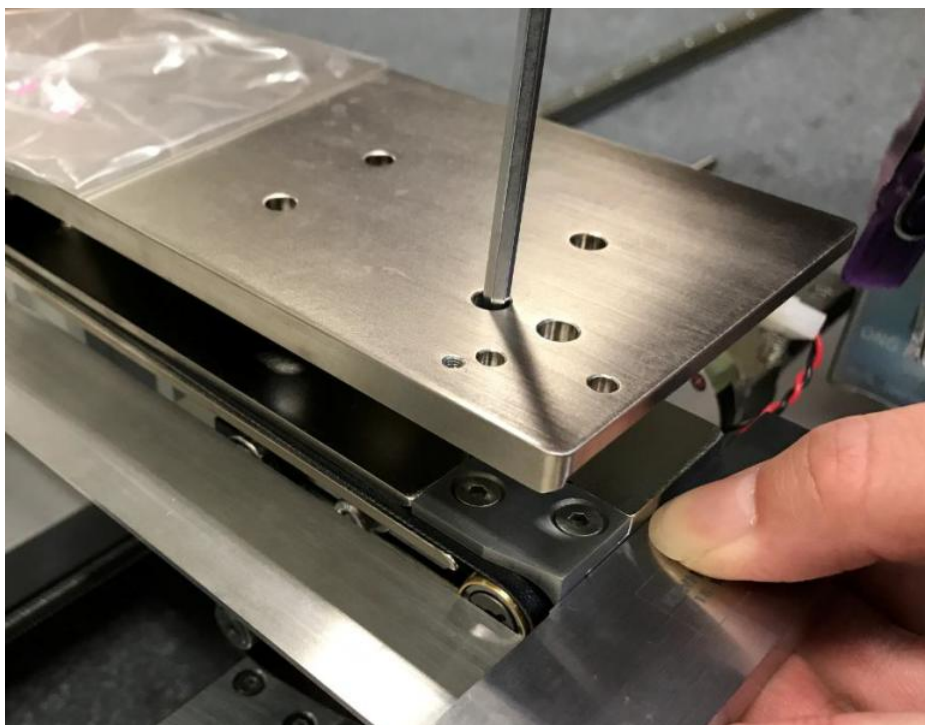


Figure 44 Adj Rail Mount Pulley Alignment

ViTrox 	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Adjustable Rail Motor Pulley & End Pulley Removal & Replacement	Effective Date	7 July 2017

- d) Toggle “Stage belt Motor ON “at Digital IO tab.
- e) Slightly loosen End Pulley assembly as below Figure 45

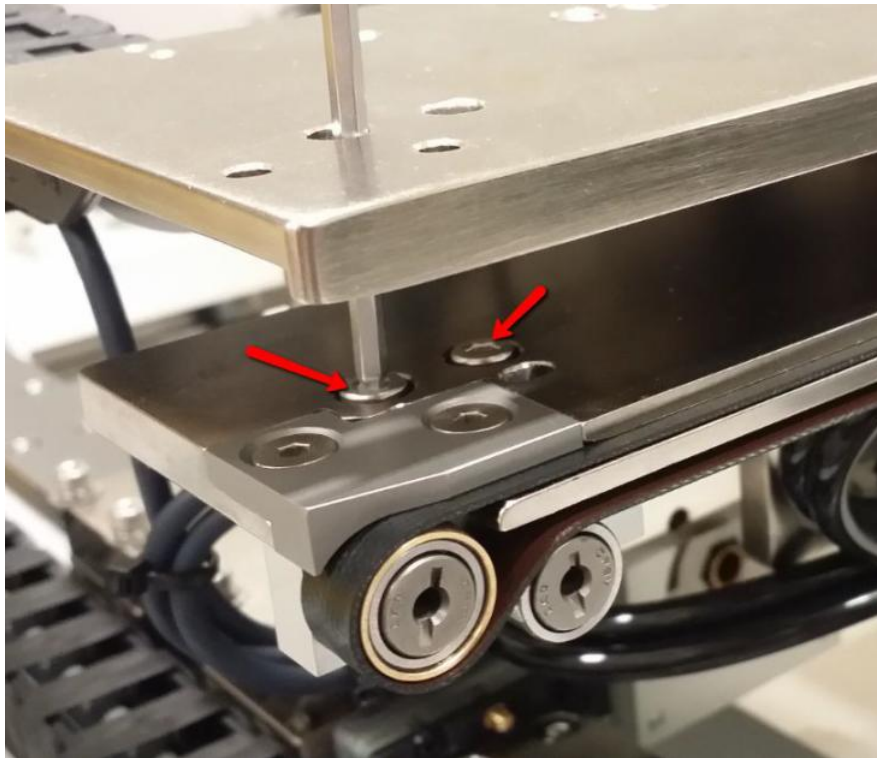


Figure 45 End Pulley Assembly

- f) Insert Allen Key to End Pulley as below Figure 46

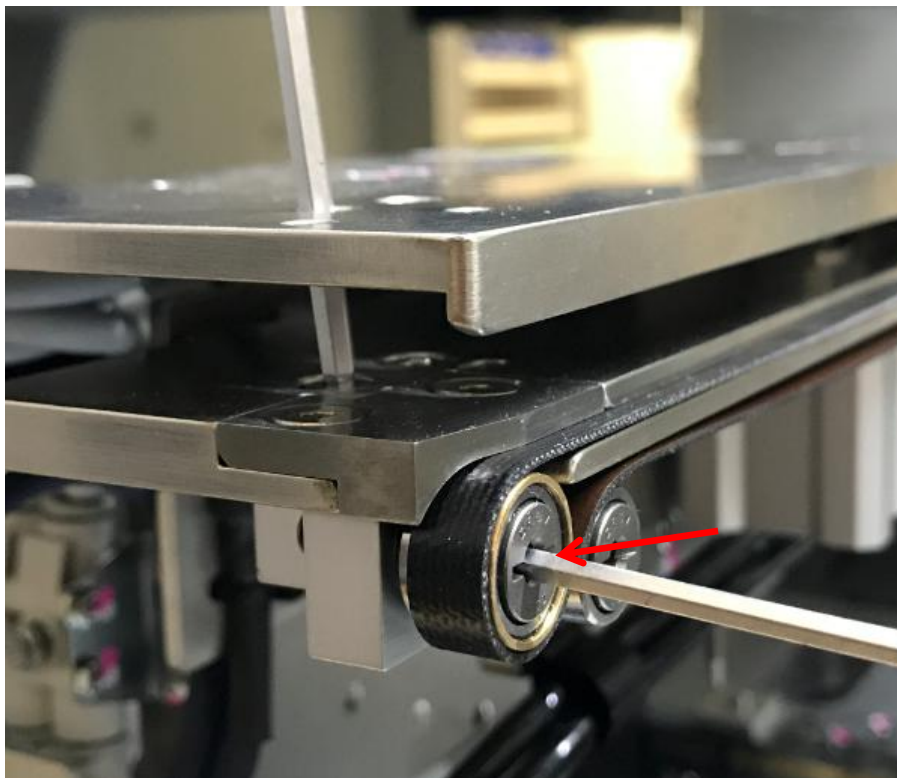


Figure 46 End Pulley Assembly

ViTrox	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Adjustable Rail Motor Pulley & End Pulley Removal & Replacement	Effective Date	7 July 2017

- g) Align the End Pulley until Belt Flow towards Inner Direction as below Figure 47. Tighten 2 x Screws after desire alignment achieve

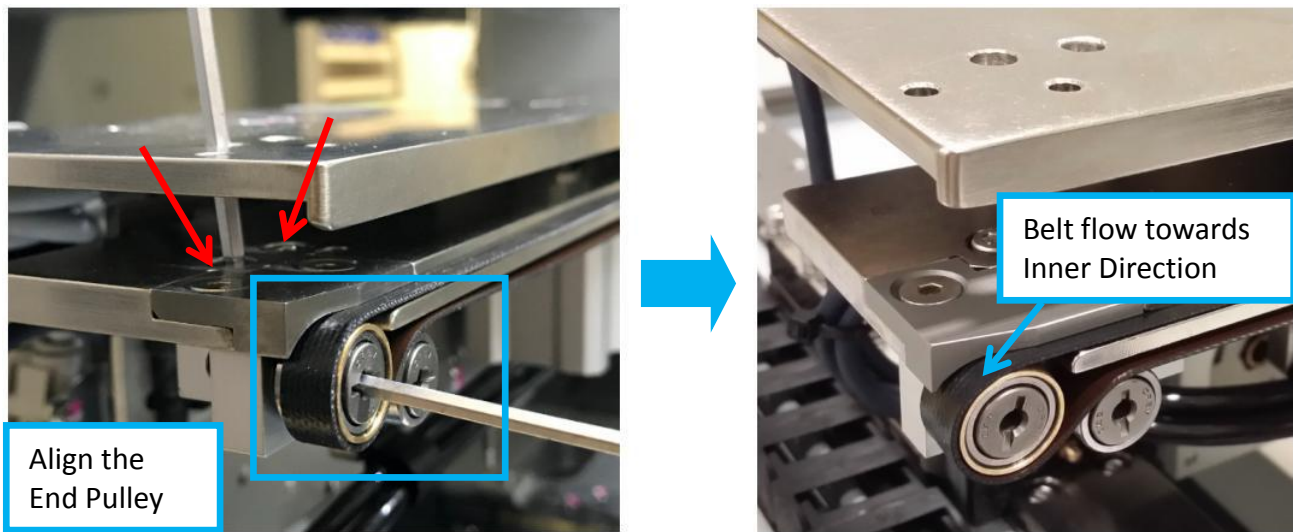


Figure 47 End Pulley Alignment

7. Toggle “Stage Belts Direction Right to Left “ while “Stage Belts On” remain ON.

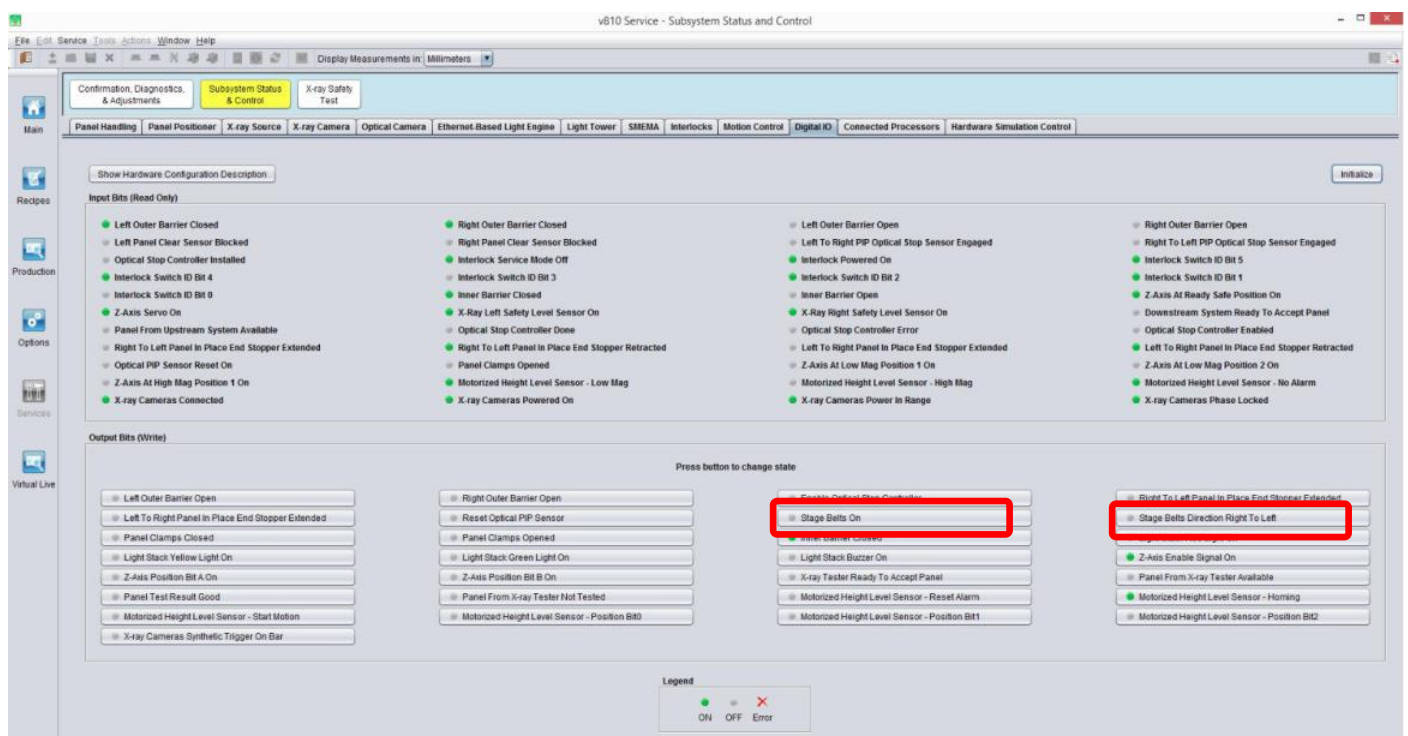


Figure 48 Digital IO (Belt Direction Switch)

8. Visual check Conveyor Belt Running Condition.
9. Repeat Step 6~8 if Conveyor Belt Running Condition fail
10. Toggle Off “Stage Belt On” at Digital IO tab
11. Toggle Off “ Stage Belt Direction Right to Left” at Digital IO tab
12. Remove all tools and close all door access